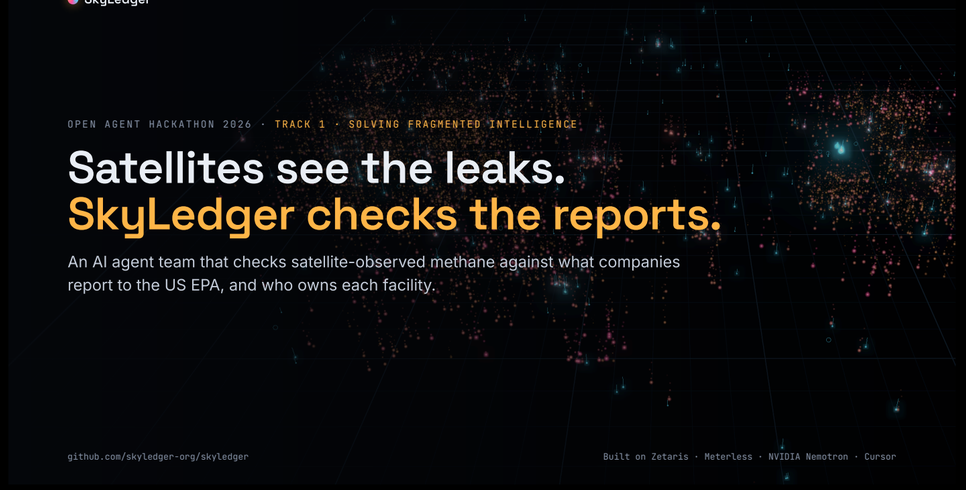
SkyLedger

OPEN AGENT HACKATHON 2026 · TRACK 1 · SOLVING FRAGMENTED INTELLIGENCE

Satellites see the leaks.
SkyLedger checks the reports.

An AI agent team that checks satellite-observed methane against what companies report to the US EPA, and who owns each facility.



github.com/skyledger-org/skyledger

Built on Zetaris · Meterless · NVIDIA Nemotron · Cursor

01 · THE PROBLEM

Three sources that were never designed to meet.

kg / hour

What satellites see

A methane plume at a point, on one morning. Carbon Mapper (Tanager, EMIT, AVIRIS-NG, GAO).

t CO_2e / year

What companies report

One annual total per facility ID to the EPA Greenhouse Gas Reporting Program.

free text

Who owns it

“TARGA RESOURCES (73%); PIONEER NATURAL RESOURCES USA INC (27%)”: inconsistent strings, no IDs.

No shared keys, no shared units, no shared clock. Nobody reconciles them at scale, so a leak seen from space rarely meets the number a company reported.

02 · WHO IT'S FOR, AND WHY NOW

The people who must judge whether a reported number is plausible.

- **ESG and credit analysts** signing off on issuers' emissions disclosures.
- **Regulators and state agencies** prioritising inspections.
- **Investigative journalists and NGOs** who need evidence, not anecdotes.

Why now

- California **SB 253**: first Scope 1 and 2 reports due **10 Nov 2026**, with no assurance requirement yet.
- EPA has **delayed RY2025 GHGRP reporting** and proposed ending most of the program. Independent checks matter more.
- Satellite methane data is now **public and frequent** (Carbon Mapper Tanager).

03 · THE SOLUTION

Every plume becomes a traced, signed-off dossier.

Find

Federated geo-join of plumes to reporting facilities, queried in place through Zetaris.

Attribute

Wind and plume geometry decide which facility sits at the plume's head. Nemotron reads the image as a check.

Reconcile

kg/h → tonnes per year, always as a range, against the facility's report and its parent companies.

Challenge

A Skeptic must rule out five rival explanations before anything is written. An analyst signs before it can be shared.

Output: draft dossiers with ranges, evidence (SQL, image, computation), a parent-company roll-up, and a hash-chained trace of every agent step. Wording is guarded: “inconsistent with available data”, never an accusation.

04 · AGENT WORKFLOW

Eight agents. One shared state. A Skeptic with veto power.



Not a linear chain

The Skeptic sends work back: widen the search (GeoLinker), look again (PlumeReader), stop annualising (Reconciler). Loops are bounded by config.

Numbers come from code

Distances, unit conversion and ratios come from SQL and code. Models interpret and word; a guard rejects any number the code didn't compute.

Traceable

Every step goes to `logs/trace-<run>.jsonl`, hash-chained so any edit breaks the chain, and streams live to the console.

05 · THE SKEPTIC

Before a word is written, five rival explanations must fall.

RIVAL	THE QUESTION	HOW IT'S TESTED (DATA, NEVER OPINION)
R1 Misattribution	Could another facility be the source?	Plume head vs. candidate distance and wind; basin-level operators nearby are flagged
R2 Intermittency	A one-off release?	Sightings inside the reporting-year window; fewer than 2 blocks any annual figure
R3 Not a reporter	Is the site required to report?	Subparts filed, CH_4 reported, or no reporter at the plume head
R4 Coverage	Does the report cover this equipment?	Basin-aggregate reports (W-ONSH/W-GB) and Subpart W coverage
R5 Data quality	Is the measurement usable?	Rate present, uncertainty below the configured limit, wind present

06 · WHAT IT FINDS · RUN EXAMPLE-1, PERMIAN BASIN (TX/NM), GHGRP RY2023

Fewer flags. Better ones.

3,413
satellite methane sources in the scan area

60
plumes analysed end to end by the agents

31
plumes with no reporting facility at the plume head

18
cases where nearest-facility matching blames a plant more than 1 km away

34
discrepancies a naive baseline would flag

→

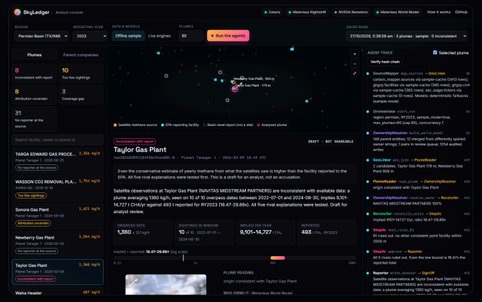
8
still standing after the Skeptic

Wrong facility blamed, on **15** hand-labelled plumes:

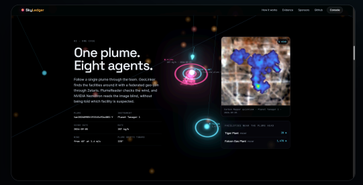
Nearest facility **33%** · Single LLM **47%** · **SkyLedger 7%**

A small sanity check labelled by the team, not expert ground truth. Details in eval/results.md.

07 · THE PRODUCT



Analyst console: live engine status, streaming agent trace, basin map, plain-language dossier, sign-off.



Landing page: the real plume and facility data in WebGL, following one plume through the agents.



08 · TECHNICAL STACK

Small, reproducible, honest about what's live.

- **TypeScript on Node 22**, run with tsx. No build step.
- **HTTP API:** `POST /run`, live streaming over server-sent events, signed sign-off.
- **Front end:** vanilla JS, three.js, GSAP, Lenis, served locally. No CDN.
- **Docker:** builds and runs from a clean clone with no keys, no network and no GPU.
- **Config, not constants:** every threshold in `config/skyledger.json`, every region in `config/regions/`.
- **Providers:** `AI_PROVIDER=nightshift | nvidia | none`.
- **Sample mode:** cached public data (Carbon Mapper, EPA GHGRP RY2023) and deterministic agents. Identical results every run.
- **Data:** Carbon Mapper (non-commercial terms), US EPA GHGRP, SEC EDGAR.

09 · HOW WE USED THE SPONSOR TECHNOLOGY

SPONSOR	ROLE IN SKYLEDGER	PROOF
Zetaris	The federated geo-join: Carbon Mapper sources × GHGRP facilities × CH_4 totals, queried in place over the REST SQL API from three Lightning filestore tables	Live result identical to the cache for 60/60 plumes; SQL and row hash in every dossier
Meterless Nightshift	Primary AI provider, built to the official provider contract (headers, discovery, token handling, Auto Detect)	Provider done-criteria pass; Nemotron replies with receipts
Meterless World Model	Facility → owner → parent graph: merges messy owner strings, provenance on every write, review queue	Conformance suite 8/8 against the engine we run
Meterless Scout Intent	Analyst sign-off issues an execution contract in Scout's shape (HMAC over canonical JSON)	Contract binds the dossier hash and the trace head
NVIDIA Nemotron	3.5 Lightning writes each dossier via Nightshift; 3.5 Super VL reads each plume image blind	Vision measured honestly and kept advisory (below)
Cursor	Development environment (see CURSOR.md)	

Unique

- **Data never leaves its source:** Zetaris joins satellite and regulatory data in place, and the same SQL is shown to the analyst.
- **A World Model for corporate ownership:** owner strings resolved into parents with provenance, the gap Zetaris's own docs name.
- **Agents that check agents:** the Skeptic blocks and sends work back, and the trace proves it.
- **Real data as the interface:** every number on the landing page and in the deck comes from the API.

Measured, not claimed

- **Vision:** blind Nemotron headings agreed with the wind on 24 of 64 live readings (38%, about 33% by chance) on 114-pixel quicklooks. So vision is advisory: it can add confidence, never downgrade a case alone.
- **Route disclosure:** Nightshift 1.2.7 refuses image requests to Nemotron VL. PlumeReader tries Nightshift first, then calls NVIDIA directly, logging each fallback. Reported to Meterless.

11 · FUTURE ENHANCEMENTS

From a basin to a monitoring service.

Coverage

Every US basin and every reporting year; scheduled refresh from Carbon Mapper and EPA; SEC CIK links for parents.

Accuracy

A published labelled test set and accuracy report; vision on larger scenes or as a data-quality check; H-MEM to learn attribution rules from analyst decisions.

Product

Hosted service with logins and audit trails; alerts when a facility's gap persists; export of signed dossiers for disclosure reviews.

Commercial use needs a data agreement with Carbon Mapper, and a legal review of wording that names companies. Both are part of the plan.